FREE GUIDE · LOAD AS MEDICINE

Post-arthroplasty loading guide

Getting strength back after a knee or hip replacement. An educational guide, not your rehab plan.

Before you start: your surgical team comes first

This guide is for people who have had a knee or hip replaced, or who are preparing for it. It is educational. It is not your rehab plan.

- **Use these exercises only after your surgeon or physical therapist has given you the go-ahead for them.** Every operation and every recovery is different, and the timing that suits one person can be too early for another.
- **Your team's protocol wins.** If their instructions differ from anything here, including the acceptable pain guide, follow theirs.
- **Take this guide to your next appointment** and ask which parts fit your stage of recovery.

Get help now (call 911 or your local emergency number) for:

- pain, swelling, warmth or tenderness in your calf or the back of your thigh;
- chest pain, or sudden shortness of breath.

After surgery these can be signs of a blood clot, which needs urgent care.

Contact your surgical team today if:

- the wound is red, hot, leaking or opening up;
- you feel feverish or unwell;
- the joint suddenly becomes much more swollen or painful;
- you suddenly cannot bend or straighten it as far as before.

Stop and contact a clinician if

- your calf hurts or swells after a joint replacement, or you get chest pain, sudden breathlessness, or new trouble controlling your bladder or bowels (call 911 or your local emergency number);
- the joint is hot, red or swollen and you feel feverish or unwell;
- a surgical wound is red, hot, leaking or opening up;
- the joint looks out of shape or cannot take weight after a fall or injury;
- the joint starts giving way, or catching and getting stuck so you can't fully bend or straighten it, or this gets worse;
- you suddenly cannot bend or straighten it as far as usual;
- pain at rest or at night does not ease when you change position, most nights for two weeks or more;
- new numbness, tingling or weakness appears in your leg;
- you lose about 10 pounds or more without trying.

These signs are worth checking before you exercise. They are not a diagnosis.

What one review of trials found

After a knee replacement (total knee arthroplasty), getting strength back in the leg is a big part of recovery. A review of 14 trials, covering 880 people after knee replacement, found that resistance exercise for the legs (exercise against a weight, band or machine) led to somewhat better walking speed, a slightly faster sit-stand-walk test (about 1 second), more leg strength and a little more knee bend (about 3 degrees) than standard exercise, measured right after the programme. The overall quality of the evidence was low. PROMISING Wei 2024

Your surgical team decides when and how this fits your recovery.

Resistance exercise compared with standard exercise, right after the programme

PROMISING

What was measured	What the review found
Walking speed (fastest pace)	Somewhat faster
Sit-stand-walk test	About 1 second faster
Leg strength (straightening and bending the knee)	Stronger
Knee bend	About 3 degrees more
Six-minute walk, stair climbing, straightening the knee	No clear difference

Low-quality evidence; the effects were small and measured only right after the programme. Source: Wei 2024. One review of trials. Ask your surgical team how resistance exercise fits your recovery.

How much ache is acceptable?

When a joint hurts, it is natural to read every ache as a sign of harm. Pain is your body's alarm system. With arthritis, a moderate ache during exercise does not by itself mean the joint is being damaged. PROMISING Bedson 2008; Bricca 2019; Fransen 2015 Physical therapists often use a simple guideline to help people keep moving without guessing.

Rate the ache from 0 to 10. 0 is no pain at all. 10 is the worst pain you can imagine.

0	1	2	3	4	5	6	7	8	9	10

During the activity: up to about 5 out of 10. Many clinicians treat an ache that stays moderate, at about 5 or below, as acceptable during exercise. If it climbs above 5, ease off: go slower, use less weight, or stop that exercise for today.

By the next day: back to your usual. Your usual is how the joint normally feels on an ordinary day. Any extra ache should settle back to that within about 24 hours. If it has, that amount of activity was a dose your joint recovered from.

This is clinical guidance, not a tested safety limit for arthritis. UNPROVEN-BRIDGE

If it has not settled after 24 hours, do less next time. Shorten the session, lower the weight, or do fewer sets, and build back up once the joint is back to its usual. One harder day is

information about the dose, not a sign you should stop moving.

When the guideline does not apply

- **After surgery, your surgeon's or physical therapist's protocol comes first.** Use the pain levels and limits they give you, even if they differ from these.
- **A hot, swollen joint, or one that catches and gets stuck so you can't fully bend or straighten it, is not an "acceptable ache".** Stop and see a clinician (see the red-flag box).
- **Sharp, sudden or a new kind of pain** is a reason to stop that activity, even if it scores below 5.
- **Rheumatoid or other inflammatory arthritis:** this guideline was written with osteoarthritis in mind. Ask your rheumatology team whether it fits you.
- **It does not replace advice from a clinician who knows your history.** If they have told you something different, follow them.

Questions to take to your next appointment

- When can I start exercise against a weight, band or machine?
- Which exercises fit my stage of recovery right now?
- What level of ache is acceptable for me at this stage?
- Which signs mean I should call you, and which number should I use?

Notes from my appointment

About the evidence in this guide

Every research statement here carries a grade:

- ESTABLISHED consistent evidence from human studies or guidelines.
- PROMISING good early evidence in people, with limits noted.
- UNPROVEN-BRIDGE a reasoned link or clinical guidance not yet tested directly.

Sources

- Wei G, Shang Z, Li Y, Wu Y, Zhang L. Effects of lower-limb active resistance exercise on mobility, physical function, knee strength and pain intensity in patients with total knee arthroplasty: a systematic review and meta-analysis. BMC Musculoskelet Disord. 2024;25(1):730. PMID 39267026.
- Bedson J, Croft PR. The discordance between clinical and radiographic knee osteoarthritis: a systematic search and summary of the literature. BMC Musculoskelet Disord. 2008;9:116. PMID 18764949.
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- Fransen M, McConnell S, Harmer AR, et al. Exercise for osteoarthritis of the knee. Cochrane Database Syst Rev. 2015;1:CD004376. PMID 25569281.

Medical disclaimer

This guide is educational. It does not provide medical advice, diagnosis or treatment, and it is not a substitute for evaluation by a qualified clinician. Talk with your physician or physical therapist before beginning a new exercise program, particularly after a recent surgery. After joint surgery, your surgeon's or physical therapist's protocol comes first. Stop any session for chest pain, dizziness, or unusual shortness of breath, and see the red-flag box at the front of this guide for signs that need a clinician.

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